From Understanding to Intervention: Towards an Agenda for Countering Dark Patterns in Games

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Abstract. Human-Computer Interaction (HCI) research has significantly advanced our understanding of Deceptive or Dark Patterns (DPs) in digital games through identification, ethical analysis, and awareness initiatives. However, a critical gap persists in developing robust, empirically validated methods to counteract these manipulative designs effectively. Prevailing business models and financial incentives often lead to the perception that proposing countermeasures is unrealistic despite widespread concerns about fairness and exploitation. This position paper directly challenges this assumption, arguing that moving beyond foundational understanding towards applied, interventionist research is crucial to bridge the 'implementation gap' between ethical insight and design practice. We call for expanding HCI games research and outline an agenda that prioritizes developing player-centric interventions informed by rigorous evidence and creating actionable design strategies for practitioners. Such initiatives are essential not only for mitigating DPs harms but also for demonstrating the feasibility of designing for genuine player well-being alongside sustainable developer goals, thereby paving a path toward ethical game design.

Keywords: Digital Games · Video Games · Games · Game Design · Games Research · Dark Patterns · Deceptive Design · Deceptive Patterns · Deceptive Design Patterns · Manipulative Patterns · Manipulative Design · Deceptive Design Practices · Malicious Design · Manipulation · Ethical Game Design · Ethical Design · Design Ethics · Ethics · Responsible Design · Awareness

1 Introduction

The pervasive use of Deceptive or Dark Patterns (DPs) in online gaming—i.e., design approaches crafted to shape player behavior, frequently for commercial reasons—has drawn considerable critical attention in the Human-Computer Interaction (HCI) field [13,15,35]. HCI researchers have made significant contributions in uncovering and classifying these trends [13], exploring their consequences to players [9,18,32], drawing their implications for ethics [16], and bring-

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ing attention to them through pedagogical efforts [7,38]. Even with advancements toward understanding DPs in games, we still lack tested, empirically supported ways to counteract them effectively. Little HCI-focused, empirically tested interventions or design alternatives exist that are as easy to implement or can be simply integrated into workflows as DPs currently are.

Indeed, some have argued that critically examining and proposing countermeasures to these patterns is an impractical and futile effort, given their perceived alignment with prevailing business models and financial incentives within the games industry [1,29]. This perspective emphasizes the documented conflicts: video game developers express the strain of economic pressures, specifically in games-as-a-service models, while simultaneously recognizing the ethical issues surrounding game elements they commonly use (for example, loot boxes) [1,5]. Furthermore, attempts at top-down regulation face significant hurdles; for instance, Belgium's 'ban' on paid loot boxes has been reported as largely ineffective due to poor industry compliance, enforcement challenges linked to funding and scale, and the ease with which players can circumvent technical blocks [40,41]. This contrasts with player attitudes, where strong concerns about fairness, transparency, exploitation, and manipulation are prevalent [9,31], alongside broad consensus on the need for mitigation policies [31].

This complex and somewhat contradictory scenario is echoed in conferences, workshops, and collaborative projects, where many fellow games and HCI researchers and practitioners applaud the attempts at actively countering deceptive design practices for their moral value but deem them ultimately unrealistic. Although anecdotic evidence, this type of perception reflects (and is possibly reinforced by) the real-world regulatory struggles [40,41], the documented tension between developer constraints and perceived market demands [1], and the only moderate feasibility for several proposed mitigation strategies, such as consumer protection regulations or industry self-regulation [5,6].

This position paper directly challenges that pragmatic (or perhaps resigned) perspective. We consider that this oftentimes underlying assumption prematurely closes off opportunities for necessary and actionable design innovations in this field. The critical question then becomes not *if* we can counter dark patterns in games, but *how* HCI and game researchers can effectively lead this task in such a way that practitioners can readily incorporate their findings and proposals in their game development workflow.

While acknowledging the economic realities, we believe that a sustained emphasis on identification and awareness, as important as these are, is not enough to effect real change. The pursuit of ethical game design, rather than a utopian exercise in conflict with industry sustainability, can enhance player engagement, build deeper trust, and ultimately support a more sustainable gaming environment. However, to realize this potential, there is an actual "implementation gap" [23] between ethical insight and design practice that must be bridged. It is precisely this gap – the need for robust, tested, and game-specific countermeasures – that HCI and games research should spearhead.

Therefore, we call not for a shift but a systematic expansion of HCI games research to dedicate significant effort towards more interventionist and practical methodologies, complementing the necessary ongoing work in identifying and analyzing DPs in games. This paper shares reflections on current practices and visions for the community's future, as well as research on games' DPs. We highlight the need to prioritize the systematic evaluation of targeted countermeasures, the empirically guided design of player-centered intervention instruments, and the development of further implementable design strategies that practitioners can directly apply. Such efforts are needed to reduce the harms of DPs in games and demonstrate the feasibility and validity of designing for actual player well-being as a primary ethical consideration while also exploring its conjunction with sustainable developer goals.

2 How HCI Has Been Approaching Dark Patterns in Games

HCI researchers have increasingly addressed DPs in the last few years. Almost since the coining of the term "Dark Pattern" in 2010, HCI scholars and practitioners have been central to this discourse, and the prominence and volume of research on this topic have only risen, especially since 2021 [13,15]. HCI research has significantly advanced our understanding of DPs in digital games by focusing on understanding and detecting these patterns and their effects on users and developing ethical frameworks to protect players from manipulative design strategies.

2.1 Identification and Classification Work

Research on DPs has mainly focused on their classification and detection [28]. Zagal, Björk, and Lewis presented in 2013 one of the earliest and most influential works to define DPs in games and categorize different types [42]. Since then, HCI researchers have spent considerable efforts detecting and classifying more DPs or supporting existing taxonomies and typologies. For instance, Niknejad et al. [29] conducted a large-scale study on nearly 1,500 mobile games, using Zagal and colleagues' taxonomy as a basis for this analysis. Their study highlighted how these DPs exploit players through temporal, monetary, social, and psychological means.

Another example is the Dark Pattern Analysis Framework (DPAF) [28], which categorizes 68 types of DPs and provides game-specific examples, adding 20 DPs types in comparison to a recent taxonomy by Gray et al. [14]. Other HCI studies classify DPs in games in both traditional and emerging forms, such as augmented and virtual reality, focusing on how game features can intensify their impact on players [1,19]. Researchers also examine the interplay between cognitive biases and DPs, identifying when users are most susceptible to manipulation and suggesting ethical interventions [26].

The classification field has advanced significantly over the past decade, shifting from simple taxonomies to more detailed categories considering digital platforms and cognitive biases. However, efforts to create a universally accepted taxonomy for DPs continue. Detection methods have also progressed from manual to semi-automated or fully automated approaches [24]; nonetheless, studies reveal that current tools only detect 45% of known patterns [29], highlighting the need for more robust detection models.

2.2 Awareness and Education Initiatives

HCI research has proposed mitigation strategies, such as regulatory support, ethical design guidelines, and user protection mechanisms, to reduce the harm caused by DPs in games [12,19,26]. There are also educational initiatives aimed at expanding awareness of DPs among designers, developers, and users and integrating these topics into design and computing curricula to foster more responsible and ethically informed design practices.

For example, Iantorno and McKee [17] argue for embedding critical discussions of deceptive design into HCI and design education, proposing pedagogical frameworks that make space for ethical reflection and resistance strategies. Similarly, Gray et al. [12] call for including dark pattern literacy in UX education, suggesting that formal training can prepare future designers to identify and avoid manipulative design practices. Efforts like these aim not only to inform but also to instill professional responsibility and critical engagement with the ethical dimensions of interface design.

Researchers have also explored serious and persuasive games as educational countermeasures for users [1,20]. Such games aim to inform and foster behavioral change through immersive, interactive experiences. Unlike static awareness tools, these games strive to engage players in a way that mirrors real-world manipulation, making them a promising HCI-based approach for educating users about deceptive design in games.

However, research has increasingly highlighted the limitations of simply raising user awareness to combat the effects of deceptive patterns in interfaces. While initial studies focused on informing users about the existence of manipulative designs, emerging evidence shows that awareness by itself often falls short [4,20]. For instance, Bongard-Blanchy and colleagues [4] observed that even users who are aware of manipulative strategies often feel incapable of opposing them, signaling the need for more diverse and effective interventions that go beyond just placing responsibility on users.

2.3 Ethical, Legal, and Health Considerations

HCI research has critically examined how DPs can distort player experience, undermining autonomy, trust, and long-term satisfaction. Scholars have advocated for user-first game design principles that foreground transparency, informed consent, and user agency [1,26]. These critiques highlight the ethical concerns surrounding manipulative mechanics – such as exploiting cognitive

biases or guilt-tripping players – to drive engagement or monetization at the expense of player well-being [11].

Additionally, legal and policy-oriented research within HCI has begun shaping public discourse by framing DPs as unfair commercial practices. This type of work contributes to regulatory efforts that seek to align design with consumer protection laws [15,25]. Interdisciplinary workshops and position papers have further advanced this agenda, calling for collaborative action across ethics, law, health, and design domains to address the broader societal harms posed by deceptive game mechanics [35,37].

While ethical, legal, and health-focused research has been essential in exposing the harms of DPs in games, it is not sufficient on its own to counter them effectively. Ethical critiques often remain theoretical and lack mechanisms for enforcement, while regulatory approaches are slow to adapt and challenging to apply consistently across jurisdictions [15,25]. Health-oriented research has noted adverse psychological effects, such as compulsive play and diminished well-being [3], but these findings rarely lead to enforceable standards. Besides, industry pressures and monetization often undermine ethical intentions, making it hard for designers to resist manipulative mechanics [1]. Without integrated strategies that include education, design accountability, and better policy coordination, these efforts struggle to challenge the commercial logic sustaining DPs in games.

Taken together, this body of work suggests that HCI research is moving toward increasingly multidisciplinary approaches, drawing on design, law, psychology, and ethics to tackle the challenges posed by deceptive patterns in games. While taxonomies and critical analyses have provided valuable conceptual foundations, concrete and efficacious countermeasures remain scarce. Addressing this gap requires an expanded, more interventionist, and practice-oriented approach – e.g., one that incorporates player advocacy into design processes, fosters cross-sector collaboration among HCI researchers, policymakers, and industry stakeholders, and advances the development of real-time detection mechanisms and transparency tools for players.

3 A Call To Action: Why (and How) We Can (and Should) Go Beyond Our Current Efforts

HCI, by definition, sits at the crossroads of human psychology, design, and technology. This position affords a particular advantage in addressing DPs in games, not by imposing ethical dictates or making policy alone, but by analyzing how interactive systems can better serve both players and developers. HCI researchers are specialists in finding out about users—in this case, players. We find out about players' needs, contexts, motivations, cognitive processes, and psychological needs, such as autonomy, competence, and relatedness, through methods like ethnographic research, playtesting, experiments, and player experience studies, to name a few perspectives of our work. The empirical understanding that this type of research produces is the foundation. Ethicists can tell us what harm is, and legislators can legislate against it, but HCI researchers and practitioners

can look at how to counteract it in practice. HCI can investigate **not only the nature and impact of DPs but also the intrinsic nature of particular design choices** on fundamental human needs like autonomy and informed consent, viewing these as rights to be upheld within the interactive experience.

Hence, we as a community can focus on **how to fight DPs practically in games**. While it is valuable to discuss ethical game design principles [16] or game design patterns [27], HCI can take these concepts further. We can make, build, and test physical interfaces and interaction techniques that **actively promote** player autonomy and well-being. And this should not have to be at the expense of the game industry's business goals. For instance, HCI researchers are equipped to investigate *in what way* a game's progression system may be portrayed to communicate effort vs. reward without exploitative grinding, or in-game purchase offers may be presented to honor informed choice rather than exploiting urgency. HCI provides the tools to both formulate and empirically test such alternatives.

The task is not so much to eliminate monetization or engagement mechanics but to design them sustainably and ethically. HCI can take the lead with novel interfaces and interaction techniques that aim for genuine player engagement through enjoyment and intrinsic motivation rather than compulsion. For example, if a game is highly compelling and treats player agency with respect, it has a greater likelihood of developing long-term loyalty and favorable community attitudes that are also business assets. HCI can examine and demonstrate *how* designs rooted in respect for player well-being can foster genuine engagement, which in turn may supplement or even redefine sustainable business objectives rather than simply optimizing for pre-existing ones. For example, systems that provide players with explicit control of their playtime and data or interfaces that overtly disclose monetization mechanics could be a space for HCI-led innovation.

Therefore, our task is not to replace the fundamental work of ethicists, lawyers, psychologists, or teachers. Instead, we call for HCI research to *produce* concrete strategies and tools informed by empirical evidence upon which those other disciplines can rely. We can show *what* is effective at the interface level to avert harm and strengthen positive player experience. This evidence-based, pragmatic, design-centered approach, with a focus on the design of the interaction itself, is HCI's distinctive and core contribution to addressing DPs in games. We must aim to truly equip developers with well-informed tools that allow them to create compelling games people enjoy without resorting to behaviors undermining their autonomy and well-being. Building on this view, the following subsections indicate practical alternatives and tangible outputs we envision to foster a game development ecosystem that prioritizes player autonomy through empirically validated ethical design practices.

3.1 Collaboration to Document Positive Patterns and Share Working Examples of Ethical Game Design Elements

Recent studies, such as those by Miranda and coauthors [27], propose alternative design patterns called "radiant game design patterns" to mitigate DPs and satisfy

player psychological needs. Lewis [22] similarly suggests "motivational design patterns" while also warning against DPs that cause long-term motivational harm. Aagaard and colleagues [1] delineate the origins of DPs in mobile game development, use, and commercialization and advocate for game experiences that prioritize player and industry professionals' well-being. However, as valuable as they are, these concepts and guidelines are not often incorporated by the game industry for various reasons, one being that conceptualizations like these remain primarily descriptive, lacking readily actionable translations.

To bridge this gap and to advance initiatives like these, **a collaborative effort** in designing, prototyping, and empirically evaluating is necessary. This effort would involve demonstrating *how* ethical, "radiant" or "motivational" and other positive design ideals translate into concrete, context-specific practices that incorporate player dignity and informed choice. For instance, Frommel & Mandryk [10] found that daily quest rewards, while engaging, can negatively impact players through fear of missing out or feeling like a chore. Based on their insights, the community could develop, test, and document alternative reward systems. And then, sharing these vetted examples *grounded in empirical findings and clear design rationales* to easily accessible channels, including but not limited to formal repositories. The idea is to ease the bridge between academic knowledge and general industry use.

Before establishing cooperative collections or databases, a common ground for documenting and sharing these ethical game design elements requires fostering a shared understanding and robust methodologies. A key element is the collaborative creation of a rich lexicon for describing problematic patterns (based on well-founded taxonomies) and their ethical counterparts—designs that respect player agency as a primary concern. This requires ongoing dialogue within the HCI and games research community, facilitated through diverse collaborative platforms such as workshops, dedicated tracks at conferences, special interest groups, and open online forums for discussion and co-creation of documentation.

Standardized yet flexible ways to validate "working examples" are also necessary to fulfill this vision. As research groups worldwide will pursue various research questions, they would benefit from shared principles for empirical verification, possibly including metrics related to both player well-being and engagement. For instance, the "System Darkness Scale" [30] represents an interest in quantifiable evaluations, offering a potential starting point, which could be adapted and combined with mixed-method approaches to capture the positive dimensions of ethical designs. The emphasis here is on demonstrating respect for the player, not solely optimizing engagement metrics.

While a dynamic, web-based repository – perhaps inspired by collaborative platforms or open-source documentation projects rather than just static sites owned by a particular researcher – remains a valuable long-term goal for structured sharing, the immediate focus should be on *diverse modes of collaboration and dissemination*. That would ideally allow researchers to contribute comprehensive documentation of design patterns. This documentation should go beyond

visual layouts and interaction sequences. It should also explicitly articulate the ethical principles and psychological constructs addressed alongside robust empirical validation. Peer review, community-led curation including developers, and versioning would be essential for maintaining this shared knowledge's quality, integrity, and evolving nature.

To encourage broad participation in these collaborative documentation and sharing efforts, the processes must be inclusive and offer clear value to contributors, such as opportunities for co-authorship, peer recognition, and fostering a community of practice around ethical game design. These shared resources can also serve as powerful educational tools, complementing efforts to raise awareness on the impact of DPs, like serious games developed with this aim [7,36,38], by providing concrete examples of ethical design in action.

Funding and sustaining such collaborative endeavors will probably represent significant challenges. Support could come from research consortia, university partnerships, or grants from foundations promoting ethical technology and open knowledge. Most importantly, sustained community engagement and shared ownership are essential for these shared resources' ongoing and long-term development, relevance, and ethical stewardship.

The decentralized nature of research can be valuable here. Various groups can contribute expertise on specific game genres, diverse player populations (e.g., children or those with particular vulnerabilities), or nuanced ethical issues (e.g., data privacy, fair monetization). Well-orchestrated collaborative efforts and thoughtfully designed sharing platforms can consolidate these diverse insights, **making specialized knowledge and practical examples of ethical design accessible** to the broader game development community, thereby promoting a design ethos grounded in respect for the player.

Despite the ambitious scope, fostering collaboration for documenting and sharing ethical design knowledge aligns with the core tenets of scholarly research. It directly addresses the demand for actionable solutions to ethical challenges in game design, as highlighted by Aagaard et al. [1]. Such collaborative efforts, focused on creating and disseminating practical, ethically grounded knowledge, can significantly accelerate the adoption of design principles that genuinely prioritize player well-being within the game industry.

3.2 Development of Software Artifacts to Foster Real-Time Awareness on Users and Developers

Good gains have been achieved in enhancing player sensitivity and insight into manipulative practices by developing and piloting serious games [7,21,36,38,39] and developing tools that enable users to spot manipulative design elements [2,7]. To provide but one illustration, Fiedler and colleagues [7] showed that their serious game significantly improves the ability of participants to identify DPs, and Kronhardt and coauthors [21] showed how game-adapted DPs integrated into a serious game can raise awareness through immediate consequences.

While raising awareness is necessary because it provides the cognitive tools for detection, HCI research can augment this impact by designing interactive

software artifacts with real-time support and intervention. Extending the learning possibilities of such projects, we propose a focused research effort on two types of software artifacts. First, *player-assisting tools*, such as browser extensions or in-game overlays. Their main goal would be to help transfer the pattern recognition skills games into immediate, contextual guidance. Such tools might highlight potential DPs as they appear, explain their manipulative mechanisms (drawing on the typologies and user reactions), and perhaps even suggest alternative actions derived from a validated repository.

The advantage provided herein is the immediate empowerment of players within their interaction, upholding their right to informed choice, and alleviating the sense of being "deceived" [38]. That could potentially enable players to bypass choices that could harm their autonomy. However, these tools are not supposed to trick the user into 'making a better choice.' That would also hurt their autonomy. Moreover, as Xiao and Denoo discussed, advanced players or game platforms might devise methods to circumvent or compromise the efficacy of such tools [40]. Instead, these mechanisms should aim to enhance player agency by fostering real-time awareness, not to diminish it by dictating choices. The core intention is to transform potentially deceptive or manipulative encounters into opportunities for informed decision-making.

There are specific concerns to consider. These tools must avoid alert fatigue from excessive notifications and monitor subtle DPs across gaming interfaces without generating high false positives. Additionally, they need to minimize being seen as intrusive to maintain a positive gaming experience.

The second type would be **developer-supported tools** for major game engines, which might proactively integrate ethical considerations into the development process. The goal here is to support developer objectives that align with player well-being. Plugging in the research on the root of DPs in design development and the issues designers encounter, such tools could help identify potentially problematic design decisions at development time and help designers brainstorm healthier alternatives. They could check against predefined DPs definitions derived from the vast literature that describes them and suggest ethically informed alternatives from the proposed repertoire of different proposals of positive design patterns.

The primary benefit is proactive, encouraging inherently ethical practices and support for designers working towards creating healthier experiences, particularly when they might not have clear guidance or face strict time limits [23]. This approach may help developers better consider, negotiate, and make informed design decisions on how to be more ethically engaged in their workflows. Of course, there are potential challenges to be considered. They include the difficulty of developing intelligent and context-aware tools that make practical proposals to players instead of just giving trivial notifications. Also, developers risk working around or dismissing such tools if they are perceived as creativity limitations, too limiting in design choices, or conflict with perceived financial objectives.

Of course, these tools' successful adoption and efficacy would be influenced by the industry's willingness to embrace ethical design standards and integrate such support. However, this integration is a necessary embodiment of ethical principles rather than a strategy to enhance tool efficacy. It requires a broader shift towards embedding these standards deeply within development cultures. Besides, developing such software products demands an insightful grasp of player psychology founded on research surrounding cognitive bias and DPs and the pragmatics of game development. Rigorous empirical testing will also be essential to guarantee that such tools are effective and easy to use, balance the empowerment of both end-users and developers' sustainable goals, and foster a healthy interactive experience.

It is important to note that while designing developer-support tools is crucial for fostering ethical game design practices, it is equally essential to recognize the structural constraints within the game industry that may limit their adoption and impact. Many monetization-related decisions are typically made not by designers or developers but by higher-level stakeholders such as shareholders, executives, and product or monetization teams. These decision-makers often have limited or no background in game development and primarily rely on key performance indicators (KPIs) like player retention, session length, active user base, and revenue metrics to drive design directives. As a result, developers and designers may have little autonomy to reject manipulative practices, even when they are aware of the ethical concerns surrounding DPs. Given this context, we advocate for further empirical research – such as large-scale surveys or interviews with industry professionals, especially within the mobile and live-ops sectors – to better understand developers' awareness of DPs, their perceived agency in decision-making processes, and the organizational pressures they face. This line of inquiry could help clarify how deeply knowledge and decision power regarding DPs are distributed across different roles in game companies, potentially guiding more targeted and feasible counter-strategies in the future.

3.3 Advancing Methods and Measures for Assessing Dark Pattern Prevalence and Player Experience, Including Well-Being

Current HCI research excels at identifying and categorizing DPs, although not as much within games, and provides critical explorations of their ethical and psychological impacts [16]. Significant steps have been made toward evaluation tools, such as the System Darkness Scale [30] that assess a system's perceived "darkness," and frameworks like the ADD framework [8] for critically considering DPs in children's applications. While these analytical insights and identification methods are foundational, we advocate for expanding efforts to develop and validate robust methods and measures that can more **directly and quantifiably assess the impact of specific game design elements** on player autonomy, perceived manipulation, and holistic psychological well-being.

The aim is not merely to identify whether well-being is affected but to understand and quantify this relationship with greater nuance, particularly when evaluating the efficacy of proposed countermeasures. It involves developing or

adapting specific player experience instruments, such as validated questionnaires or carefully designed behavioral metrics, that explicitly capture core tenets of ethical interaction. These instruments should focus on perceived autonomy, fairness, transparency, the sense of being respected (or manipulated), and critical aspects of psychological well-being, drawing inspiration from well-established theories like the Self-Determination Theory [33,34]. Such measures would allow researchers to move beyond general assessments of "darkness" to indicate how specific patterns or their alternatives influence these fundamental human needs, a necessary point of departure to investigate how to counteract them.

Building upon existing frameworks, we call for creating standardized yet adaptable methodologies for evaluating both the presence and, crucially, the impact of DPs across diverse game genres and monetization models. It includes refining our approaches to measure the actual effect of interventions designed to mitigate DPs or enhance player awareness. For example, understanding why some educational interventions show limited impact on knowledge about DPs matters [39] requires more sensitive evaluation protocols to determine true efficacy and identify areas for improvement.

Addressing these challenges involves **not only identifying the absence of DPs but also evaluating the effectiveness of ethical design choices** in creating a positive, engaging game experience driven by genuine enjoyment, not coercion. We can enhance our use of gameplay data by developing telemetry and behavioral analytics to uncover patterns that indicate compromised player autonomy or manipulative mechanics. Integrating this data with qualitative insights and refined player experience tools can lead to a more comprehensive assessment. Additionally, we could improve methodologies for assessing the prevalence of DPs in new games by combining expert evaluations tailored to specific game types and monetization strategies, along with emerging automated detection approaches where ethically appropriate.

Developing and adapting methods and validated measures is not an end in itself. Instead, it is a step toward generating highly specific, empirically grounded guidelines that practitioners can trust and readily apply. These methods will allow us to move from general advice to concrete, evidence-backed recommendations (e.g., "*To uphold player autonomy in reward systems, design X which demonstrably leads to Y outcome in perceived fairness, as measured by Z*"). By focusing on quantifiable impacts on core aspects of ethical player experience, we can make assessments more systematic, comparable across studies, and ultimately more impactful in driving real change in game design practices.

4 Forging Ahead: A Research Agenda

This position paper argued for systematically expanding the focus of HCI games research regarding game DPs. It means moving from identifying what is problematic to demonstrating what is possible. We urge a dedicated focus on interventionist and practical methodologies that enrich and are enriched by our existing foundational understanding of DPs, creating a robust cycle of theory and application. The aim is not to offer definitive, one-size-fits-all solutions, as the landscape

of game design and player experience is far too diverse and dynamic for such prescriptions. Instead, we have intended to outline a call to action, highlighting specific areas where research efforts can make a tangible difference in fostering more ethical and player-respecting game environments.

From our discussion, a research agenda emerges, embracing the complexities and committing to developing empirically grounded, actionable knowledge by:

- **Developing and Validating Ethical Solutions:** How can the HCI community collaboratively document, prototype, and rigorously evaluate specific positive patterns or ethical design alternatives for familiar game mechanics prone to exploitation? What methodologies best capture their impact on player autonomy, well-being, and potential alignment with sustainable developer goals?
- **Creating and Assessing Intervention Tools:** What types of player-assisting software artifacts (e.g., real-time awareness tools, contextual guides) can effectively empower players against DPs without being overly intrusive or easily avoided? Conversely, how can we design developer-support tools to seamlessly integrate ethical considerations into existing game development workflows, offering practical guidance rather than mere warnings? What are the most effective ways to empirically test the real-world impact of such tools?
- **Refining Methods for Impact Assessment:** How can we develop more nuanced and validated metrics and methodologies to assess the prevalence of DPs and their tangible impact on player well-being and perceived fairness? Can we create standardized yet flexible protocols for evaluating the effectiveness of countermeasures across diverse game genres and player populations?
- **Translating Research into Actionable Guidance for Practitioners:** What are the most effective formats and channels for translating empirical findings into practical, context-specific design guidelines and heuristics that game developers can readily understand and apply? How can we better bridge the implementation gap between academic research and industry practice?

Addressing these questions requires a multi-faceted approach, integrating qualitative depth with quantitative rigor and fostering stronger collaborations between researchers, designers, developers, and players. More importantly, it should come from people from different backgrounds and countries, including the Global South perspective. This endeavor is not about dictating what games should be but expanding their palette – engaging, entertaining, and fundamentally respectful of the individuals who play them. This call to action is an invitation to the HCI and games research community. By focusing our collective expertise on these interventionist and practical trajectories rooted in theory, we can move beyond critique to creation, spearheading the development of knowledge and tools that genuinely support designing a more ethical future for digital games.

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References

1. Aagaard, J., Knudsen, M.E.C., Bækgaard, P., Doherty, K.: A game of dark patterns: designing healthy, highly-engaging mobile games. In: CHI Conference on Human Factors in Computing Systems Extended Abstracts, pp. 1–8 (2022)
2. Baroni, L.A., Pereira, R.: Deceptive patterns under a sociotechnical view. In: Proceedings of the XXIII Brazilian Symposium on Human Factors in Computing Systems, IHC 2024. Association for Computing Machinery, New York (2024). https://doi.org/10.1145/3702038.3702081
3. Birk, M.V., Van Der Hof, S., Hodent, C., Gerling, K., Van Rooij, A.J.: Behavioural design in video games: ethical, legal, and health impact on players. In: Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems, pp. 1–4 (2023)
4. Bongard-Blanchy, K., Rossi, A., Rivas, S., Doublet, S., Koenig, V., Lenzini, G.: "I am definitely manipulated, even when i am aware of it. It's ridiculous!" - dark patterns from the end-user perspective. In: Proceedings of the 2021 ACM Designing Interactive Systems Conference, DIS 2021, pp. 763–776. Association for Computing Machinery, New York (2021). https://doi.org/10.1145/3461778.3462086
5. Brock, T., Johnson, M.: The gamblification of digital games (2021)
6. Colder Carras, M., Carras, M., Labrique, A.B.: Stakeholders' consensus on strategies for self-and other-regulation of video game play: a mixed methods study. Int. J. Environ. Res. Public Health **17**(11), 3846 (2020)
7. Fiedler, K., Schäfer, R., Borchers, J., Röpke, R.: "Deception Detected!"—a serious game about detecting dark patterns. In: Schönbohm, A., et al. (eds.) GALA 2024. LNCS, vol. 15348, pp. 191–200. Springer, Cham (2024). https://doi.org/10.1007/978-3-031-78269-5_18. Chapter 18. The LNCS volume is formally part of the 2025 series, though published in late 2024
8. Fitton, D., Read, J.C.: Creating a framework to support the critical consideration of dark design aspects in free-to-play apps. In: Proceedings of the 18th ACM International Conference on Interaction Design and Children, pp. 407–418 (2019)
9. Freeman, G., Wu, K., Nower, N., Wohn, D.Y.: Pay to win or pay to cheat: how players of competitive online games perceive fairness of in-game purchases. Proc. ACM Hum.-Comput. Interact. **6**(CHI PLAY), 1–24 (2022)
10. Frommel, J., Mandryk, R.L.: Daily quests or daily pests? The benefits and pitfalls of engagement rewards in games. Proc. ACM Hum.-Comput. Interact. **6**(CHI PLAY), 1–23 (2022)
11. Gray, C.M., et al.: Mobilizing research and regulatory action on dark patterns and deceptive design practices. In: Extended Abstracts of the CHI Conference on Human Factors in Computing Systems, pp. 1–6 (2024)
12. Gray, C.M., Kou, Y., Battles, B., Hoggatt, J., Toombs, A.L.: The dark (patterns) side of UX design. In: Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems, pp. 1–14 (2018)

13. Gray, C.M., Sanchez Chamorro, L., Obi, I., Duane, J.N.: Mapping the landscape of dark patterns scholarship: a systematic literature review. In: Companion Publication of the 2023 ACM Designing Interactive Systems Conference, DIS 2023, pp. 188–193. Companion, Association for Computing Machinery, New York (2023).https://doi.org/10.1145/3563703.3596635
14. Gray, C.M., Santos, C., Bielova, N.: Towards a preliminary ontology of dark patterns knowledge. In: Extended abstracts of the 2023 CHI Conference on Human Factors in Computing Systems, pp. 1–9 (2023)
15. Gray, C.M., Santos, C.T., Bielova, N., Mildner, T.: An ontology of dark patterns knowledge: foundations, definitions, and a pathway for shared knowledge-building. In: Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems, CHI 2024. Association for Computing Machinery, New York (2024). https://doi.org/10.1145/3613904.3642436
16. Hodent, C., Blumberg, F., Deterding, S.: Ethical games: toward evidence-based guidance for safeguarding players and developers (2024)
17. Iantorno, M., Guadagnolo, D., Petterson, A.: Dark patterns and pedagogy: expanding scholarship and curriculum on manipulative marketing practices. AoIR Selected Papers of Internet Research (2023)
18. King, J., Fitton, D., Cassidy, B.: Investigating players' perceptions of deceptive design practices within a 3d gameplay context. Proc. ACM Hum.-Comput. Interact. **7**(CHI PLAY) (2023). https://doi.org/10.1145/3611053
19. Krauss, V., et al.: What makes XR dark? Examining emerging dark patterns in augmented and virtual reality through expert co-design. ACM Trans. Comput.-Hum. Interact. **31**(3), 1–39 (2024)
20. Kronhardt, K., Rolfes, K., Gerken, J.: Trickery: exploring a serious game approach to raise awareness of deceptive patterns. In: Proceedings of the International Conference on Mobile and Ubiquitous Multimedia, pp. 133–147 (2024)
21. Kronhardt, K., Rolfes, K., Gerken, J.: Trickery: exploring a serious game approach to raise awareness of deceptive patterns. In: Proceedings of the International Conference on Mobile and Ubiquitous Multimedia, MUM 2024, pp. 133–147. Association for Computing Machinery, New York (2024). https://doi.org/10.1145/3701571.3701588
22. Lewis, C.: Motivational design patterns. Ph.D. thesis, UC Santa Cruz (2013)
23. Lindberg, S., Karlström, P., Männikkö Barbutiu, S.: Cultivating ethics–a perspective from practice. In: Proceedings of the 11th Nordic Conference on Human-Computer Interaction: Shaping Experiences, Shaping Society, pp. 1–11 (2020)
24. Mathur, A., et al.: Dark patterns at scale: findings from a crawl of 11k shopping websites. Proc. ACM Hum.-Comput. Interact. **3**(CSCW) (2019). https://doi.org/10.1145/3359183
25. Mathur, A., Kshirsagar, M., Mayer, J.: What makes a dark pattern... dark? Design attributes, normative considerations, and measurement methods. In: Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems, CHI 2021. Association for Computing Machinery, New York (2021). https://doi.org/10.1145/3411764.3445610
26. Mildner, T., Inkoom, A., Malaka, R., Niess, J.: Hell is paved with good intentions: the intricate relationship between cognitive biases and dark patterns. arXiv preprint arXiv:2405.07378 (2024)
27. Miranda, D.M., Pontes, R.M., de Gois Ribeiro Darin, T.: It's dark but just a game: towards an ethical and healthy game design practice. In: Proceedings of the 21st Brazilian Symposium on Human Factors in Computing Systems, pp. 1–7 (2022)

28. Nie, L., Zhao, Y., Li, C., Luo, X., Liu, Y.: Shadows in the interface: a comprehensive study on dark patterns. Proc. ACM Softw. Eng. **1**(FSE), 204–225 (2024)
29. Niknejad, S., Mildner, T., Zargham, N., Putze, S., Malaka, R.: Level up or game over: exploring how dark patterns shape mobile games. In: Proceedings of the International Conference on Mobile and Ubiquitous Multimedia, pp. 148–156 (2024)
30. van Nimwegen, C., Bergman, K., Akdag, A.: Shedding light on assessing dark patterns: Introducing the system darkness scale (SDS). In: 35th International BCS Human-Computer Interaction Conference, pp. 1–10. BCS Learning & Development (2022)
31. Petrovskaya, E., Zendle, D.: Predatory monetisation? A categorisation of unfair, misleading and aggressive monetisation techniques in digital games from the player perspective. J. Bus. Ethics, 1065–1081 (2022). https://doi.org/10.1007/s10551-021-04970-6
32. Petrovskaya, E., Zendle, D.: "These people had taken advantage of me": A grounded theory of problematic consequences of player interaction with mobile games perceived as "designed to drive spending". Hum. Behav. Emerg. Technol. **2022**(1), 1260174 (2022)
33. Ryan, R.M., Deci, E.L.: Self-determination theory and the facilitation of intrinsic motivation, social development, and well-being. Am. Psychol. **55**(1), 68 (2000)
34. Ryan, R.M., Deci, E.L.: Self-determination Theory: Basic Psychological Needs in Motivation, Development, and Wellness. Guilford publications (2017)
35. Sánchez Chamorro, L., Bongard-Blanchy, K., Koenig, V.: Ethical tensions in UX design practice: exploring the fine line between persuasion and manipulation in online interfaces. In: Proceedings of the 2023 ACM Designing Interactive Systems Conference, DIS 2023, pp. 2408–2422. Association for Computing Machinery, New York (2023). https://doi.org/10.1145/3563657.3596013
36. Santos Filho, L., Sousa Junior, A., Ribeiro, G., Darin, T.: They say that the world was built for two but deceptive patterns say otherwise: a critical game for player empowerment. In: Proceedings of the XXIII Brazilian Symposium on Human Factors in Computing Systems, IHC 2024. Association for Computing Machinery, New York (2024). https://doi.org/10.1145/3702038.3702076
37. Sas, M., Denoo, M., Mühlberg, J.T.: Informing children about privacy: a review and assessment of age-appropriate information designs in kids-oriented F2P video games. Proc. ACM Hum.-Comput. Interact. **7**(CHI PLAY) (2023). https://doi.org/10.1145/3611036
38. Tjostheim, I.: A serious game about apps, data-sharing and deceptive design. In: Guarda, T., Portela, F., Diaz-Nafria, J.M. (eds.) Advanced Research in Technologies, Information, Innovation and Sustainability, vol. 1936, pp. 332–343. Springer, Cham (2024). https://doi.org/10.1007/978-3-031-48855-9_25
39. Tjostheim, I., Ayres-Pereira, V., Wales, C., Manna, A., Egenfeldt-Nielsen, S.: Dark pattern: a serious game for learning about the dangers of sharing data. In: Proceedings of the 16th European Conference on Games Based Learning, vol. 16. Academic Conferences International Limited, Reading (2022). https://doi.org/10.34190/ecgbl.16.1.872. Section: Late Submissions
40. Xiao, L.Y.: Breaking ban: Belgium's ineffective gambling law regulation of video game loot boxes. Collabra Psychol. **9**(1), 57641 (2023)
41. Xiao, L.Y., Denoo, M.: What did players think about Belgium's 'ban'on loot boxes? In: Abstract Proceedings of DiGRA 2023 Conference: Limits and Margins of Games (2023)
42. Zagal, J.P., Björk, S., Lewis, C.: Dark patterns in the design of games. In: Foundations of Digital Games 2013 (2013)